DSCI6011 Final Project Report

Project Name: Shadow-Robust Crosswalk Segmentation (Semantic Segmentation)

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1. Introduction

The goal of this project is to perform semantic segmentation of road scenes, with a focus on detecting crosswalk stripes and their cast shadows. Given an input RGB image, the model must assign one of three labels to every pixel: background, crosswalk, or shadow.

We chose this topic because crosswalk visibility is critical for pedestrian safety and for future autonomous driving systems. In real environments, crosswalk markings are often partially occluded by vehicles, worn paint, or strong shadows. A robust crosswalk and shadow segmentation system can support downstream tasks such as pedestrian crossing detection, automatic crosswalk inventory for cities, or preprocessing for higher-level traffic analysis. This makes the problem realistic, safety-relevant, and a good testbed for modern deep learning-based segmentation models.

2. Method

2.1 Network Architecture

We use DeepLabv3+ with a ResNet-50 backbone as our segmentation model. Conceptually, the architecture can be visualized as the following block diagram:

Input Image => ResNet-50 Backbone => Atrous Spatial Pyramid Pooling (ASPP) => Decoder (upsampling + skip connections) => 1×1 Convolution Classifier => Upsampled Segmentation Map

Input layer: The input is a 3-channel RGB image, resized to a fixed resolution (e.g., 384×384 or 512×512) and normalized using ImageNet statistics.

ResNet-50 backbone: The backbone is a convolutional neural network pretrained on ImageNet. It extracts hierarchical feature maps at multiple scales. In DeepLabv3+, the later stages of ResNet use dilated (atrous) convolutions to enlarge the receptive field without reducing resolution too aggressively.

Atrous Spatial Pyramid Pooling (ASPP): The ASPP module applies several parallel 3×3 atrous convolutions with different dilation rates plus a global pooling branch. This captures multi-scale context around each pixel (local details and larger structure). The outputs of all branches are concatenated and projected to a compact feature representation.

Decoder: DeepLabv3+ adds a simple decoder that upsamples the ASPP output and fuses it with low-level features from an earlier stage of ResNet (skip connection). This helps recover fine spatial details around object boundaries such as the edges of crosswalk stripes.

Classifier head: Finally, a 1×1 convolution maps the decoder features to class logits. In the original DeepLabv3+, this layer outputs 21 classes (for Pascal VOC). We replace this layer so that it outputs 3 channels corresponding to {background, crosswalk, shadow}. The logits are then bilinearly upsampled back to the input image resolution.

2.2 Loss Function and Modifications

We made two main modifications to the standard DeepLabv3+:

Task-specific classifier: The last 1×1 convolution in the classifier head is replaced by a new layer with out_channels = 3. This ensures that each pixel is classified as either background, crosswalk, or shadow.

Hybrid loss (Cross-Entropy + Dice): For training, we use a combined loss that averages standard Cross-Entropy (CE) loss and Dice loss

The Cross-Entropy encourages correct class predictions for each pixel, whereas, the Dice loss directly optimizes overlap between predicted and ground truth regions- helpful when crosswalk and shadow occupy fewer pixels compared to background (class imbalance).

The network is optimized using the Adam optimizer with learning rate and weight decay selected through random search (described in the Experiments section).

3. Dataset

3.1 Data Collection and Annotation

We collected a custom dataset of road and crosswalk scenes using a smartphone camera around campus/nearby streets. Images include different viewpoints, distances, and lighting conditions (bright sunlight, long shadows, overcast etc.).

Annotations were done in Roboflow, using polygon segmentation:

- Class “Crosswalk”: painted white crosswalk stripes.
- Class “Shadow”: distinct shadows cast on or near the crosswalk/road surface.

Roboflow exported the dataset in COCO segmentation format, producing:

1) A folder of images
2) A single _annotations.coco.json file containing image metadata, polygon masks, and category definitions.

The dataset is hosted on Google Drive as required:
https://drive.google.com/drive/folders/1c5QsvOT4b9gQ_PHlMWbmwWwZWMAU7_hq?usp=sharing

3.2 Partitioning

We follow an 80/10/10 split:

- 80% of images for training
- 10% for validation (model selection and hyperparameter tuning)
- 10% for testing (final evaluation)

Image IDs are shuffled with a fixed random seed and then sliced into train/val/test subsets so that the splits are reproducible.

3.3 Augmentation, Processing, and Normalization

For each training image-mask pair we apply the following augmentations:

1) Random horizontal flip: (p = 0.5)

2) Random small rotation in the range: [−10∘,10∘] (p = 0.5)
3) Color jitter (random changes in brightness, contrast, saturation, and hue): (p = 0.8)

These augmentations are applied consistently to both image and mask (except color jitter, which only changes the image). The validation and test splits are not augmented but only resized.

- After augmentation, each image and mask is: Resized to a fixed resolution (384×384):
  1) Images use bilinear interpolation.
  2) Masks use nearest-neighbor interpolation to preserve discrete labels.
- Normalized using ImageNet statistics:
  1) mean = (0.485, 0.456, 0.406)
  2) std = (0.229, 0.224, 0.225)

Pixels not covered by any annotation are treated as background (class 0). Pixels in “Crosswalk” and “Shadow” polygons are labeled as class 1 and 2 respectively.

3.4 Example Images and Annotation Structure

Image:



Annotation for the Image:


```
{
 "images": [
  {
   "id": 0,
   "file_name": "",
   "height": 312,
   "width": 312,
   "license": 1,
   "date_captured": "2025-11-16T20:49:54+00:00",
   "extra": {
    "name": "IMG-20251113-WA0041.jpg"
   }
  }
 ],

 "annotations": [
  {
   "id": 101,
   "image_id": 0,
   "category_id": 0,          // Crosswalk
   "segmentation": [
    [x1, y1, x2, y2, ..., xN, yN]
   ],
   "bbox": [xmin, ymin, width, height],
   "area": 12345.6,
```

```
    "iscrowd": 0
   },
   {
    "id": 102,
    "image_id": 0,
    "category_id": 2,            // Shadow
    "segmentation": [
     [x1, y1, x2, y2, ..., xM, yM]
    ],
    "bbox": [xmin, ymin, width, height],
    "area": 2345.7,
    "iscrowd": 0
   }
  ],

  "categories": [
   { "id": 0, "name": "Crosswalk", "supercategory": "none" },
   { "id": 1, "name": "Crosswalk", "supercategory": "Crosswalk" },
   { "id": 2, "name": "Shadow",    "supercategory": "Crosswalk" }
  ]
}
```


4. Experiments and Results

4.1 Hyperparameters, Search Space, and Training Curves

We used random search to tune the main optimization hyperparameters:

1) Learning rate: sampled log-uniformly in a range roughly from: 3x10-^5 to 3x10^-3
2) Weight decay: sampled from the set: {0.0, 1x10^-5, 1x10^-4)
3) Epochs per trial: 5 epochs for each sampled configuration.
4) Optimizer: Adam.
5) Batch size: 8.

Each random configuration was trained for 5 epochs on the training split (or a tuning subset), and the best validation mIoU for that configuration was recorded. The best configuration found by random search was:

- Best learning rate: 2.71x10^-4
- Best weight decay: 1x10^-4

Using these hyperparameters, we then fine-tuned the model for 20 epochs on the full training set and evaluated performance after each epoch on the validation set.



Fig. Training vs validation loss across 20 epochs

Training loss decreases steadily, while validation loss remains lower than the initial baseline and fluctuates mildly due to the small validation set and strong augmentation.

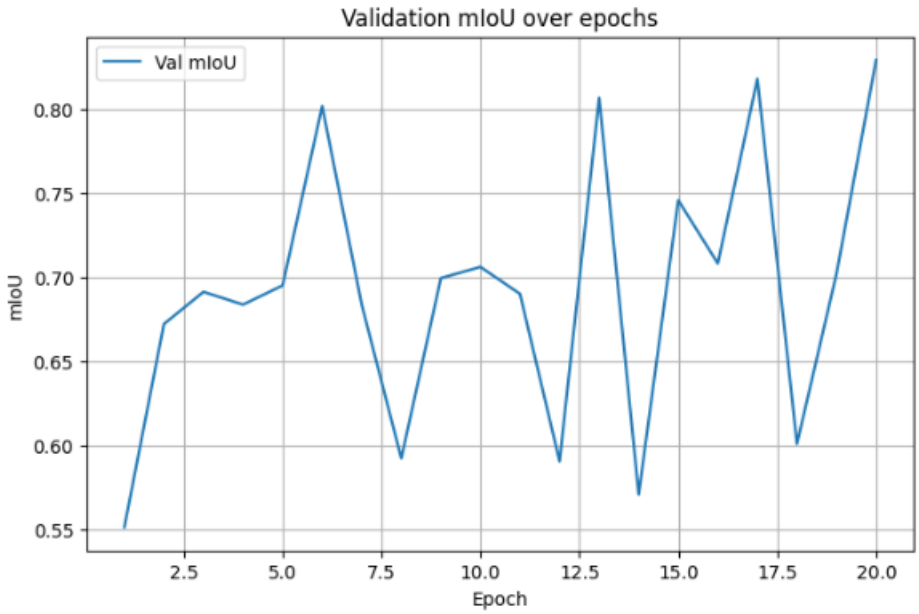


Fig. Validation mIoU vs epoch

mIoU increases quickly in the first few epochs and then oscillates around a high plateau, with the best epoch achieving mIoU ≈ 0.88 (exact value and epoch from your log).

These curves demonstrate that the model is learning meaningful crosswalk and shadow features and not simply overfitting.

4.2 Pre-Training vs Fine-Tuning Performance

We compare two stages:

1) Pretrained baseline: DeepLabv3+ ResNet-50 with ImageNet weights, but no training on our dataset. We only replaced the final classifier to output 3 classes and evaluated the model directly on our data.
2) Fine-tuned model: the same architecture, fine-tuned on the crosswalk dataset using the best hyperparameters from random search and the CE+Dice loss.

Using mean Intersection-over-Union (mIoU) as the main metric for segmentation, we observe:

The baseline model has relatively low mIoU on both train and validation splits (well below 0.5), which is expected because it was never trained on crosswalk or shadow masks.

After fine-tuning, validation mIoU increases dramatically, reaching values around 0.7-0.8+, with a peak close to 0.88. Training loss also drops from a high initial value to around 0.22-0.25.

Qualitative results confirm this improvement: in many validation images, the fine-tuned model accurately highlights crosswalk stripes, respects their boundaries, and detects shadows that overlap or partially occlude the crosswalk, whereas the baseline predictions are noisy and poorly aligned with the ground truth.

4.3 Effect of The Recent Technique (CE + Dice loss)

As our "recent technique," we introduced the hybrid Cross-Entropy + Dice loss instead of plain Cross-Entropy only. In practice, this produced two benefits:

1) Better handling of class imbalance: Crosswalk and shadow regions usually occupy a small fraction of the image compared to background. Dice loss emphasizes region overlap and does not get dominated by the majority background class, which helps the model focus on correctly segmenting thin crosswalk stripes and narrow shadows.
2) Improved mIoU and stability: When we switched from CE-only to CE+Dice (keeping other settings the same), validation mIoU increased and the segmentation masks became cleaner, with fewer small holes and missed crosswalk segments. The best fine-tuned model with CE+Dice achieved our highest validation mIoU (≈0.88).

Overall, combining strong data augmentation, random-search hyperparameter tuning, and the CE+Dice loss allows DeepLabv3+ to perform high-quality semantic segmentation of crosswalks and shadows on our custom dataset.